**Project Design Phase**
**Solution Architecture**

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| Date | 4 April 2026 |
| Team ID | SWTID-2026-3550 |
| Project Name | Streamlining Ticket Assignment For Efficient Support Operations |
| Maximum Marks | 4 Marks |

# Solution Architecture Overview

**The proposed solution uses a 3-tier architecture to automate ticket assignment and improve support operations. It connects users, application logic, and data storage through a structured and scalable system.**

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# Architecture Components

## 1. Presentation Layer (User Interface)

- **Web-based interface for:**
  - **Customers (to raise tickets & track status)**
  - **Support team (to manage tickets)**
  - **Admin (to configure system)**
- **Technologies: HTML, CSS, JavaScript**

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## 2. Application Layer (Backend Logic)

**Core Modules:**

- **Authentication Module: Handles login and access control**
- **Ticket Management Module: Creates and manages tickets**
- **Routing Engine: Automatically assigns tickets based on category, roles, and groups**
- **Workflow Engine: Manages ticket lifecycle (Open → In Progress → Closed)**
- **Notification Module: Sends alerts and updates**

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## 3. Data Layer (Database)

**Data Stored:**

- **User details**
- **Ticket information**
- **Roles and groups**

- **Workflow configurations**

**Technology: Relational Database / ServiceNow Tables**

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## Architecture Flow (Step-by-Step)

1. **User submits a ticket through the UI**
2. **Ticket is stored in the database**
3. **System classifies the ticket based on issue type**
4. **Routing engine assigns ticket to appropriate group (e.g., Certificates / Platform)**
5. **Support team receives and processes the ticket**
6. **Status updates are sent to the user**
7. **Admin monitors and configures system workflows**

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## Key Features of the Architecture

- **Automation: Eliminates manual ticket assignment**
- **Role-Based Access: Ensures secure and controlled access**
- **Scalability: Can handle increasing number of users and tickets**
- **Reliability: Reduces errors in ticket routing**
- **Real-Time Processing: Faster ticket assignment and updates**

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## Conclusion

**This solution architecture effectively bridges the gap between business problems and technical implementation. By integrating automation, workflow management, and role-based access control, it ensures efficient, scalable, and reliable support operations.**